

Kai Luzniak

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TECHNICAL SKILLS

Languages: Python, C, C++, TypeScript, JavaScript, C#, SQL, HTML/CSS

Frameworks: React, Vite, NgRx, Node.js, Express, Astro, Unity, Mirror

Tools: PostgreSQL, Docker, AWS EC2, Cloudflare, MQTT, Git, Linux/Unix, Agile, Jira

EXPERIENCE

Northrop Grumman – Student Software Developer (HIP)

Sep 2023 – Jun 2025

- Developed embedded software in C++ and Python on Linux for a 4-Raspberry Pi smart home, integrating I2C and GPIO sensors and actuators to automate irrigation, motion-activated lighting, and motorized entry for a simulated client.
- Led a 10-week Agile development cycle as Scrum Master, coordinating tasks and incorporating feedback across 3 rounds of user testing, then presented the completed embedded system to 200+ students and Northrop Grumman engineers.
- Programmed a voice assistant in Python, using the OpenRouter API to process speech-to-text and authenticate door-access requests, then MQTT/Mosquitto over TCP/IP to trigger a Raspberry Pi-controlled door lock.

Boeing Defense, Space & Security – Technical Program Management Intern

Jun 2024 – Aug 2024

- Developed a project management tool in HCL Leap that surfaced Agile workflow training gaps, gathering requirements from program leadership and the data team before adoption by 100+ employees.
- Translated stakeholder reporting requirements into an Excel VBA workflow that automated data cleaning and report generation for a \$1.3M R&D initiative, saving 3+ hours of manual formatting per report and supporting leadership reviews.
- Interviewed 10+ software, mechanical, and cloud engineers to quantify software download delays, then proposed a cloud deployment model projected to reduce downtime by 70% and recover \$2M+ annually in engineering productivity.
- The proposal advanced through 3 presentation rounds to become a finalist in Boeing's LINC innovation challenge.

Slug Gaming – Software Engineer Intern

Sep 2025 – Present

- Developed a website serving a 3,700+ member collegiate gaming organization using React, JavaScript, & Tailwind CSS.
- Implemented CI/CD with GitHub Actions to streamline site builds and deployment.
- Coordinated with 11 directors and 120+ players to maintain team rosters, bios, and graphics.

Code Ninjas – Coding Instructor

Oct 2024 – Jun 2025

- Mentored 100+ K–12 students in Unity (C#) and Microsoft MakeCode; led workshops on AI fundamentals & game design.

PROJECTS

Cata (2D Multiplayer Game) | *Technical Lead, Unity (C#), AWS EC2, Mirror, Unity VCS*

- Founded and led a 6-person development team to engineer a 2D online multiplayer roguelike game in Unity (C#), winning 1st Place – “Game of the Year” out of 30+ CS developers.
- Architected a headless authoritative server using Mirror and AWS EC2 to securely synchronize real-time combat, animations, and round state for 4-player network sessions.
- Mitigated network desynchronization and optimized server traffic during live playtests by minimizing RPC payloads and implementing object pooling for high-frequency projectiles.

Compass | *Angular, TypeScript, NgRx, Firebase, SCSS, Git*

- Developed reusable Angular components using TypeScript, SCSS, and Firebase for a goal-setting web application.
- Reviewed and merged code through Git pull requests to maintain reliable app behavior across a 5-person Agile team.

EDUCATION

University of California, Santa Cruz

Expected Graduation June 2028

Bachelor of Science in Computer Science | GPA: 3.93/4.0

Activities: Tech4Good Lab Research Assistant, Gen.G Collegiate Ambassador, D1 Collegiate Overwatch Player, VSA Intern